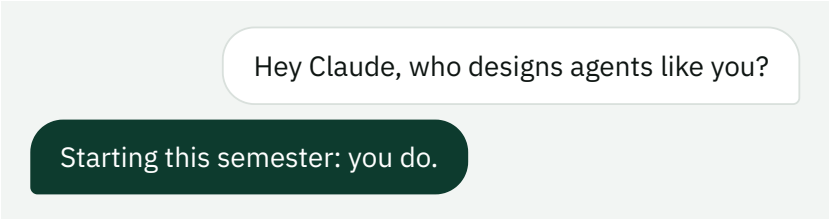
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STUDENT WORKBOOK, PART 1 · WINTER TERM 2026/27

Conversational Agent Design Lab

Department of Information Systems · University of Münster

PART 1 · STEP E1 · V2 · 20.10.2026

TEAM _____

NAME _____

TOPIC _____

HOW TO USE THIS WORKBOOK

Seven steps, nine sessions, one chain of design knowledge

This workbook follows you through the semester. Each of the seven steps (E1 to E7) begins with an overview: what happens, what you should be able to do, the key concepts, your tasks, the deliverable, the exit check and what to read. Worksheets follow each step. Fill them in as a team; they are the raw material for your seminar paper.

> **This is part 1: step E1.** You get the next parts at the start of the next steps. Type into the boxes in any PDF reader (Adobe Acrobat Reader, Preview, Firefox, Chrome, Edge) and save the file, or print it and write by hand.

The companion site holds the same content plus the Theory Lab: small simulated agents that let you try out the theories behind your design decisions, the case marketplace and the library.

	STEP	WEEKS	SESSIONS	RESULT
E1	Problem	KW 43–44	Kickoff	D1 Problem brief
E2	Requirements	KW 44–45	Character and problem space	D2 Character, objectives and requirements
E3	Design and build 1	KW 46–47	Vibe Coding 1	D3 Prototype v1 and feature trace
E4	Demonstrate	KW 48–50	User testing, Interim presentations	D4 Proof of concept (test round 1)
E5	Design and build 2	KW 51–01	Vibe Coding Advanced	Prototype v2, deployed
E6	Evaluate	KW 02	Team work with users	D5 Proof of value (test round 2)
E7	Abstract	KW 03–05	Design Knowledge Creation Video, Industry input: agents in practice, Final presentations	D6 Design principles and final submission

What you submit

Seminar paper. WI template, about 12 pages; design principles in the anatomy format

Artifact. Working, deployed agent; every feature traced to a requirement

Workbook. Worksheets and design journal with AI transparency statement; the more individual contribution

Presentations. Interim and final presentation

The weighting is not fixed yet. We publish it here and on LearnWeb as soon as it is.

Team



Dr. Timo Strohmann, Theory, design science, kickoff; Companion case



Amilia Fröhlich, Character workshop and user testing; FinBot and VIVA cases



Hans-Henning Näscher, Vibe coding and agent development; personal finance and Second Thought cases



Dr. Rangina Ahmad, Lead AI Agent Architect, Parloa; industry input in KW 03 or 04

Ground rules

- Supervision: your case supervisor is your first contact, after each session and whenever you need it.
- Feedback: we always welcome feedback, on the sessions, the material and the site. Tell us after a session, or write to your supervisor or to Timo.
- AI tools are welcome, including for code and writing. Log what you used in your design journal and submit an AI transparency statement (W10).
- Test users are people: informed consent, and no personal data in prompts or logs you cannot protect.

SEMESTER PLAN

Sessions and deadlines

	SESSION	WHEN	OWNER	BRING
1	Kickoff	Tue 20.10.2026, 13:00 to 16:00	Timo Strohmann	Curiosity; A laptop; One agent you use or used, and one thing that annoys you about it
2	Character and problem space	KW 44 (26.10 to 30.10), day open	Amilia Fröhlich	D1 Problem brief; Notes from at least three user conversations; One quote that surprised you
3	Vibe Coding 1	KW 46 (09.11 to 13.11), day open	Hans-Henning Näscher	D2 Character, objectives and requirements; A laptop with the tools Hans announces
4	User testing	KW 48 (23.11 to 27.11), day open	Amilia Fröhlich	D3 Prototype v1, running, and your feature trace (W5); Names of possible test users
5	Interim presentations	KW 50 (07.12 to 11.12), day open	All	Your talk; A running prototype; One requirement you now doubt
6	Vibe Coding Advanced	KW 51 (14.12 to 18.12) or KW 02 (11.01 to 15.01) (week open)	Hans-Henning Näscher	Your prototype and your open technical problems
7	Design Knowledge Creation Video	KW 02 (from 11.01.2027), self-paced	Timo Strohmann	Your feature trace (W5); Your round 2 results with the reasons why (W8)
8	Industry input: agents in practice	KW 03 or KW 04 (18.01 to 26.01), day open	Rangina Ahmad (Lead AI Agent Architect, Parloa)	Your draft design principles (W9); One question about building agents in practice
9	Final presentations	Wed 27.01.2027	All	Prototype v2; Your design principles

Deadlines

Before the character workshop (KW 44)	D1 Problem brief
Before Vibe Coding 1 (KW 46)	D2 Character, objectives and requirements
Before the user test session (KW 48)	D3 Prototype v1 and feature trace
KW 50, with the interim presentation	D4 Proof of concept (test round 1)
End of KW 01 (Sun 10.01)	Prototype v2 deployed
End of KW 02 (Sun 17.01)	D5 Proof of value (test round 2)
Fri 05.02.2027	D6 Design principles and final submission
Fri 09.04.2027	WI 2027 Student Track (optional)

For now the plan shows calendar weeks (KW); the exact days follow after the kickoff. The current plan is always on the companion site.

METHOD

Design science, light: seven steps

A project is a sequence of self-contained steps (echelons, Tuunanen et al., 2024). Each step ends with a result that passed its exit check. You design and build twice and test twice. Inside a step you iterate until the check holds; if a later result shows an earlier step was wrong, you step back. The method page of the companion site shows the whole semester as a figure.

	STEP	RESULT	EXIT CHECK
E1	Problem Understand the problem	D1 Problem brief	The problem statement names no technology. Each perspective rests on at least one concrete source: a document, a paper, a user. You can say what surprised you in the user conversations.
E2	Requirements Define requirements	D2 Character, objectives and requirements	Every requirement serves at least one objective. Every requirement is traced to a user insight or a theory. No requirement prescribes a technology or a specific feature.
E3	Design and build 1 Design and build	D3 Prototype v1 and feature trace	The prototype runs end to end for your core scenario. Every feature is linked to a requirement, and every requirement has at least one feature. You can show a conversation where the agent fails, and what happens then.
E4	Demonstrate Test with users	D4 Proof of concept (test round 1)	You tested with at least four users from outside the course. Each finding is linked to a feature and a requirement. Each decision for v2 follows from a finding.
E5	Design and build 2 Design and build	Prototype v2, deployed	Every change traces back to a finding from round 1. Test users can reach the prototype without your help. Logging captures what you need for round 2.
E6	Evaluate Test with users	D5 Proof of value (test round 2)	Every requirement has a verdict with evidence. Every verdict has a reason why. The limits of your data are written down.
E7	Abstract Abstract	D6 Design principles and final submission	Every principle rests on at least one evaluated feature and its reason why. A designer outside your team understands each principle without your paper. The paper states what the principles are grounded in, and what not.

The chain of design knowledge

01 **Problem statement** E1
What does not work, for whom, and why it matters, seen from practice, research and users.

02 **Design objectives and requirements** E2
Objectives say what the agent should achieve; requirements say what it must do or have to get there.

03 **Design features** E3 to E6
The concrete, buildable realisation of each requirement: built, tested with users, revised and evaluated.

04 **Design principles** E7
Abstracted from the evaluated features: what to do, for whom, in which context, and why it works (Gregor et al., 2020).

TWO TEST ROUNDS

E4 Demonstrate: do the features work for users? Formative, proof of concept. **E6 Evaluate:** are the requirements met, and why? Summative, proof of value. The reason why is what you abstract into design principles in E7.

DESIGN JOURNAL AND AI USE

After every step and every week: what we decided, because of what, what we doubt, and which AI tools we used for what (W10). The journal is part of your workbook submission.

E1

STEP E1 · KW 43–44 · UNDERSTAND THE PROBLEM · PROBLEM STATEMENT

Problem

Whose problem is worth an agent, and how do practice, research and users see it?

At the kickoff you visit five case stations, form a pair and hand in your top two cases. Each case comes with a roughly framed gap and material: papers, practice insights, sometimes user interviews. Then you look at the problem from three perspectives and talk to first users.

SESSIONS

Kickoff · Timo Strohmann · Tue 20.10.2026, 13:00 to 16:00. Input, case gallery walk, pairs and case choice. Understand what conversational agents are, why designing them is a research problem, and how we work with design science. Visit the five case stations, form a pair and hand in your top two cases.

YOU WILL BE ABLE TO

- Explain what distinguishes chatbots, assistants, companions and agents
- Name the four theory families we build on
- Describe your case's problem from the perspective of practice, research and users
- Talk to at least three potential users

DELIVERABLE D1 · BEFORE THE CHARACTER WORKSHOP (KW 44); BY THEN YOU HAVE WORKED THROUGH YOUR CASE MATERIAL

Problem brief

One to two pages: the problem statement from the perspectives of practice, research and users; first user insights; your design question.

KEY CONCEPTS

Three perspectives. Practice: who runs such an agent and what goes wrong today. Research: what we know and what is open. Users: what people need and say. A good problem statement holds all three.

Socio-technical system. An agent is never only a model. What matters is the fit between user, task and technology (Maedche et al., 2019).

Time horizon. Ad-hoc supporter, temporary advisor or persistent companion: how long the relationship lasts changes almost every design decision (Nißen et al., 2021). Theory Lab: Time horizon.

Agency level. How much the agent decides and does on its own, and where the human stays in the loop (Holldack et al., 2026). Theory Lab: Agency dial.

TASKS

- ☐ At the kickoff: visit the case stations, form a pair and hand in your top two cases
- ☐ Read the case brief and the core papers of your case
- ☐ Try at least three existing agents in your domain
- ☐ Talk to at least three potential users (worksheet W2)
- ☐ Write the problem brief from three perspectives (worksheet W1)
- ☐ Start your design journal (worksheet W10)

WATCH OUT

- Starting from a technology (“we want to use RAG”) instead of a problem
- Only reading, never talking to users
- Assuming an agent is the right answer; sometimes a form is better

EXIT CHECK

- ☐ The problem statement names no technology.
- ☐ Each perspective rests on at least one concrete source: a document, a paper, a user.
- ☐ You can say what surprised you in the user conversations.

READING

Core. Diederich et al. (2022); Maedche et al. (2019).

Further. Schöbel et al. (2024); Følstad et al. (2021); Nißen et al. (2021).

W1 Problem brief: three perspectives

Describe the problem of your case before any solution, from the perspectives of practice, research and users. This becomes deliverable D1.

Team and case

The problem in one sentence

Who struggles with what, in which situation? No technology in this sentence.

Practice

Who runs or would run such an agent? What goes wrong today? Which rules and constraints apply?

Existing solutions we tried

At least three agents or services: what works, what fails

Research

What do we know (two to four sources)? What is still open or contested?

Users

Who are they, who else is affected, what did they tell us (W2)?

Why a conversational agent at all?

What would a form, a website or a human do better?

Our design question

How might we design an agent that … for … so that …?

W2 Interview guide and insights

Plan your first conversations with users (at least three before D1, more in step E2) and capture what you learn in a comparable way.

What we want to learn

Three research questions for the interviews

Opening questions

Context, routines, last time the problem occurred

Deepening questions

Ask for stories, not opinions: “Tell me about the last time …”

Questions about agents

Experiences with chatbots or assistants, expectations, concerns

INTERVIEW	WHO (ROLE, CONTEXT)	KEY QUOTE	NEED OR PAIN POINT	SURPRISE

WHAT COMES NEXT

After E1

Work through your case material, then hand in D1, the problem brief, before the character workshop in KW 44 (next week; the day follows on LearnWeb). Bring it and your interview notes to the workshop. Part 2 of the workbook (step E2, worksheets W3 and W4) follows before the workshop.

Keep your design journal (W10) from the start: what you decided, because of what, what you doubt, and which AI tools you used. W10 is on the companion site and in the full workbook.